
Software Test Plan (STP)

Bank Management System (BMS)

Version: 1.0

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1. Introduction

1.1 Purpose

This document defines the test plan for Bank Management System (BMS) v1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing the system described in the BMS Software Requirements Specification (SRS) and Software Architecture and Design Specification (SAD).

1.2 Scope

Testing covers customer registration and login, KYC-based account opening, cash deposits and withdrawals, intra-bank funds transfer, loan application and approval, mini-statement/passbook/receipt generation, teller-assisted counter operations, and administrative functions (user/role/branch management, reporting, audit logs). External interbank networks (NEFT, RTGS, SWIFT), physical ATM/card hardware, and the future payment gateway integration are excluded, consistent with SRS Section 1.2 and SAD Section 1.2.

1.3 References

- Bank Management System — Software Requirements Specification (SRS), Version 1.0, dated 09-09-2026, including its Requirements Traceability Matrix (Section 8).
- Bank Management System — Software Architecture and Design Specification (SAD), Version 1.0, dated 14-09-2026.
- Applicable banking/data-protection requirements and PCI-style security practices referenced in the SRS (Section 5.1) and SAD (Section 3.9).

1.4 Definitions

BMS – Bank Management System; UI – User Interface; API – Application Programming Interface; OTP – One-Time Password; KYC – Know Your Customer; DB – Database; RBAC – Role-Based Access Control; TLS/SSL – Transport Layer Security / Secure Sockets Layer; PAN – Permanent Account Number; RTM – Requirements Traceability Matrix; STP – Software Test Plan; UAT – User Acceptance Testing.

2. Test Items

- Authentication & RBAC module (registration, login, OTP, session management, logout)
- Accounts / KYC module (account opening, document capture, manager approval)
- Transactions module (cash deposit and withdrawal)
- Transfers module (intra-bank transfer, OTP for high-value transfers)
- Loans module (application submission, approval/rejection workflow)
- Reporting & statements module (mini-statement, full statement, management reports)
- Admin console (staff/user/role management, branch configuration, audit log monitoring)
- Email/SMS Gateway integration (OTP delivery, alerts, notifications)
- Central relational database (accounts, transactions, loans, audit logs)

3. Features to be Tested

Features mapped to SRS requirement IDs (see SRS Sections 4 and 5):

Requirement ID	Feature
BMSF-001	Customer registration with duplicate ID/email check
BMSF-002	Login with username/customer ID, password and OTP
BMSF-003	Account lockout after 5 failed login attempts
BMSF-004	Role-based access control (Customer, Teller, Manager, Admin)
BMSF-005	Account opening with KYC document upload
BMSF-006	KYC approval routing to Branch Manager
BMSF-007	Account activation notification
BMSF-008	Cash deposit recording and balance update
BMSF-009	Withdrawal balance and daily limit check
BMSF-010	Unique transaction receipt/reference generation
BMSF-011	Intra-bank funds transfer
BMSF-012	OTP confirmation for high-value transfers
BMSF-013	Daily transfer limit enforcement
BMSF-014	Loan application submission
BMSF-015	Loan approval/rejection with remarks
BMSF-016	Loan decision notification
BMSF-017	Mini-statement and full statement generation
BMSF-018	Staff/user account and role management
BMSF-019	Periodic report generation (transactions, accounts, loans)
BMS-NF-001	Response time $\leq 3s$ for 90% of requests (up to 200 concurrent users)
BMS-NF-002	99.5% availability during banking hours

Requirement ID	Feature
BMS-NF-003	Encryption of PII/financial data at rest and in transit
BMS-NF-004	5-year audit log retention
BMS-NF-005	WCAG 2.1 AA accessibility and responsive design
BMS-SR-001 to 005	TLS enforcement, password hashing, OTP 2FA, tamper-evident audit logs, idle-session timeout

4. Features Not to be Tested

- External interbank settlement networks (NEFT, RTGS, SWIFT) — out of scope for the current release per SRS Section 1.2.
- Physical ATM/card hardware, receipt/passbook printer drivers, and barcode/QR scanner firmware — vendor-specific, optional hardware not built as part of this project.
- Future payment gateway integration — planned only, not implemented in this release (SAD Section 3.4, 4.6).
- Underlying OS, network, and hosting infrastructure hardening — assumed handled by the hosting environment, not by the application team.

5. Test Approach / Strategy

5.1 Levels

- Unit tests — module-level tests for Auth, Accounts, Transactions, Loans, Reporting, and Admin services.
- Integration tests — Web UI ↔ API, Transaction/Transfer Service ↔ Database, Transfer Service ↔ OTP Service, Auth/Reporting ↔ Email/SMS Gateway.
- System tests — end-to-end role flows for Customer, Teller, Branch Manager, and System Administrator.
- Acceptance tests — User Acceptance Testing (UAT) with the course instructor and evaluators.

5.2 Types

- Functional testing of all BMSF-xxx requirements.
- Regression testing after each module integration.
- Performance testing against BMS-NF-001 (response time) and BMS-NF-002 (availability).
- Security testing against BMS-SR-001 to BMS-SR-005.
- Usability and accessibility testing against BMS-NF-005 (WCAG 2.1 AA, responsive design).

5.3 Entry Criteria

- A stable build of the Node.js/Express/React/PostgreSQL stack (SAD Section 3.6) deployed to the test environment.
- Seeded test data (customers, accounts, staff, sample loans) available.
- Test environment configured with TLS 1.2+ and OTP sandbox credentials.

5.4 Exit Criteria

- All High-priority functional requirements (SRS Section 6 exit criteria) implemented and verified.
- No unresolved critical or high-severity functional, NFR, or security defects.
- All test cases linked in the SRS Requirements Traceability Matrix (Section 8) executed with a Pass status.

5.5 Security Validation

- OTP validation for login and for transfers above the configurable high-value threshold.
- Password/PIN hashing verification (bcrypt) — confirm no plaintext values in the database or logs.
- Session/JWT expiry and idle-timeout behaviour (default 10 minutes).
- TLS 1.2+ enforcement — connections below TLS 1.2 must be rejected.
- RBAC boundary testing — out-of-role actions return access denied for every role.
- Account logout after 5 consecutive failed login attempts.
- Audit log integrity — logs are append-only and cannot be altered or deleted by non-admin roles.
- Input validation and fuzzing on registration, deposit/withdrawal/transfer amount, and KYC document upload fields.

6. Test Environment

6.1 Hardware

Standard development/test workstation or cloud VM for the web application and database; teller-counter hardware (receipt/passbook printer, barcode/QR scanner) tested via emulation/mockup, not physical device certification.

6.2 Software

BMS web application (React front end); Node.js + Express.js backend APIs; PostgreSQL test database; Twilio/Nodemailer sandbox for OTP and notifications; test browsers (Chrome, Edge, Firefox) on desktop and mobile viewports.

6.3 Tools

Purpose	Tool(s)
Unit testing	Jest / Mocha
API testing	Postman / Supertest
UI automation	Selenium / Playwright
Performance testing	JMeter / k6
Security scanning	OWASP ZAP
Defect tracking	Jira / GitHub Issues
DB schema & seeding	Prisma / Sequelize migration and seed scripts

6.4 Test Data

Dummy customer profiles with sample KYC documents; test savings/current accounts with varied balances and statuses; staff and admin accounts covering each role; sample loan applications in different states (Submitted, Approved, Rejected).

7. Test Schedule

Milestone	Date
Test case design	28-Sep-2026

Milestone	Date
Environment setup (staging deploy)	30-Sep-2026
Test execution start	01-Oct-2026
Test execution end	13-Oct-2026
UAT	14-Oct-2026 to 16-Oct-2026
Test summary report	18-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated), cross-referenced to the SRS RTM
- Test Scripts (Postman collections, Selenium/Playwright scripts)
- Test Data sets
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Aditi Agarwal	Prepare the test plan, coordinate execution, own the RTM, and design/execute Authentication & RBAC test cases
Test Engineer — Core Modules	Abhiroop	Design and execute Accounts/KYC, Transactions, Transfers, Loans, and Reporting/Admin test cases; lead defect triage and fixes
Test Support — Environment & Data	Aarush Khullar	Set up and maintain the test environment; prepare and refresh test data sets
Test Support — Documentation & Reporting	Amilineni Sai Dhyan Anubhav	Maintain test execution logs and compile periodic status updates and the final test summary report
Product Owner / Sign-off	Prof. Pavithra	Approve test results and sign off on readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the development sub-team
Test environment or database downtime	Maintain a backup local PostgreSQL instance for testing
OTP sandbox (Twilio/Nodemailer) rate limits or outages	Use a mock/stub OTP service for automated regression runs

Risk	Mitigation
Limited semester timeline for full regression coverage	Prioritize High-priority requirements per the SRS exit criteria before lower-priority items

11. Assumptions & Dependencies

- The Node.js + Express + React + PostgreSQL stack (SAD Section 3.6) remains stable across the testing window.
- Test data (accounts, customers, staff) is seeded and available before execution begins.
- Email/SMS gateway sandbox credentials are available for OTP-dependent test cases.
- The SRS and SAD remain the requirements and design baseline; any changes are reflected in the RTM before affected test cases are re-run.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment or database is unavailable for more than 4 hours.
- The build is unstable enough to block more than 30% of planned test cases (for example, a broken Auth module blocking all downstream flows).

Resume testing if:

- Blocking defects are resolved.
- The environment is stabilized and re-verified.

13. Test Case Management & Traceability

The Requirements Traceability Matrix in SRS Section 8 is the master mapping of requirements to test cases.

Representative examples:

Requirement	Test case(s)
BMSF-001 (Registration)	TC-Auth-01
BMSF-002 (Login + OTP)	TC-Auth-02
BMSF-003 (Account lockout)	TC-Auth-03
BMSF-008 (Deposit)	TC-Txn-01
BMSF-009 (Withdrawal limit check)	TC-Txn-02
BMSF-011 (Intra-bank transfer)	TC-Xfer-01
BMSF-012 (OTP for high-value transfer)	TC-Xfer-02
BMSF-014 (Loan application)	TC-Loan-01
BMSF-017 (Statement generation)	TC-Rep-01
BMS-NF-001 (Response time)	TC-Perf-01
BMS-SR-001 (TLS 1.2+ enforced)	TC-Sec-02

Requirement	Test case(s)
BMS-SR-003 (OTP 2FA)	TC-Sec-04

14. Test Metrics & Reporting

14.1 Metrics collected

- % of test cases executed
- % passed / failed
- Defect density and defect aging
- Requirement coverage against the SRS RTM

14.2 Reports

- Periodic execution status updates during the test-execution window
- Final Test Summary Report at the end of UAT

15. Approvals

Role	Name	Signature / Date
QA Lead	Aditi Agarwal	Aditi Agarwal. 16 th September,2026
Dev Lead	Abhiroop	Abhiroop 16 th September,2026
Product Owner	Prof. Pavithra	